

Topic 9 — Agent Collaboration and Context Passing

Coordinated Agent Sub-Workflow — Implementation & Test Documentation

Project: Hierarchical Parent–Child AI Verification Workflow

Platform: n8n

AI Model: Google Gemini

Successful Test Execution: 170

1. Assessment Objective

The workflow demonstrates a parent coordinator agent invoking a reusable child specialist workflow, passing structured context between workflows, receiving structured results, managing execution state, and producing a consolidated final decision.

2. Workflow Architecture

Parent: Run Coordinator → Initialize State → Coordinator Agent → verify_item workflow tool → Coordinator Decision Parser.

Child: When Called by Parent → Verification Specialist → Verification Result Parser.

The Parent's **verify_item** Execute Workflow Tool invokes the Child workflow and passes **item_id**, **item_type**, and **claim** using dynamic AI-supplied inputs.

3. Child Specialist Workflow

Component	Configuration / Purpose
When Called by Parent	Typed inputs: item_id, item_type, claim
Verification Specialist	Assesses trustworthiness and internal consistency
Gemini Chat Model (Child)	Temperature 0; max output tokens 1024
Verification Result Parser	Structured result: item_id, status, confidence, reason

4. Parent Coordinator Workflow

Component	Configuration / Purpose
Run Coordinator	Manual trigger
Initialize State	Creates run_id, started_at, task, and two-item batch
Coordinator Agent	Processes every item and calls verify_item once per item
Gemini Chat Model (Parent)	Temperature 0; max output tokens 2048
verify_item	Execute Workflow Tool connected to Child workflow
Coordinator Decision Parser	Consolidates run_id, counts, decision, results and summary

5. Context Passing

The Parent passes the following structured fields to the Child for each item:

item_id — unique item identifier

item_type — invoice or expense

claim — claim to be verified

Execution 170 was preceded by successful child executions demonstrating that these fields reached the Child trigger intact. The final end-to-end run successfully completed the Parent → Child → Parent chain.

6. Test Scenarios

Item	Scenario	Expected Path	Execution 170 Result
ITEM-001	Invoice total \$1,200 = \$500 + \$400 + \$300	Verified	VERIFIED
ITEM-002	\$9,999 cash expense with no receipt/approver	Flagged	FLAGGED

7. Successful End-to-End Evidence — Execution 170

Execution 170 completed successfully with the complete runtime chain. The final consolidated output was:

```
{
  "run_id": "170",
  "total_items": 2,
  "verified_count": 1,
  "flagged_count": 1,
  "overall_decision": "review",
  "results": [
    {"item_id": "ITEM-001", "status": "verified", "confidence": 1},
    {"item_id": "ITEM-002", "status": "flagged", "confidence": 1}
  ],
  "summary": "One item was verified for mathematical accuracy, while the other was flagged due to lack of documentation and potential policy violations."
}
```

8. Assessment Checklist

Requirement	Status
Child Workflow Created	✓ Completed
Child Workflow Published / Active	✓ Completed
Parent Workflow Created	✓ Completed
AI Agent Configured	✓ Completed
Chat Model Connected	✓ Completed
Temperature = 0.0	✓ Completed
Max Tokens Configured	✓ Completed
Max Iterations Configured	✓ Completed
Execute Workflow Tool Added	✓ Completed
Workflow Inputs Configured	✓ Completed
Child Workflow Connected	✓ Completed
Structured Output Parser Added	✓ Completed

Requirement	Status
State Management Configured	✓ Completed
Parent–Child Context Passing Verified	✓ Completed
Flagged Scenario Tested	✓ ITEM-002
Verified Scenario Tested	✓ ITEM-001
Final Structured Output Generated	✓ Execution 170

9. Deliverables

Recommended submission package: successful Loom/YouTube demo, public GitHub repository, Parent workflow JSON, Child workflow JSON, this documentation PDF, workflow screenshots, Execution 170 test evidence, README, and implementation notes.

10. Demo Reference

Loom demonstration: <https://www.loom.com/share/0bdf4d792b8c474faceaa59d402bf95c>

11. Source Workflow Files

This documentation reflects the uploaded n8n workflow exports for **Child - Verification Specialist** and **Parent - Coordinator Agent**, together with the successful Execution 170 result reported during validation.